IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 22 January 2025, accepted 6 February 2025, date of publication 13 February 2025, date of current version 20 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3541645

RESEARCH ARTICLE

Variable Sampling Area RRT Algorithm Based on Limiting Joint Angle for Robotic Arm Path Planning

ZHI-JIE DING[1], XIN-JIA MENG[1,2], LI-XIANG ZHANG[1,3], YI-BIN GUO[1], MENG-QIU CHEN[1], AND SHI-HANG ZHEN[1]

[1]School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan 056038, China
[2]Collaborative Innovation Center for Modern Equipment Manufacturing of Jinan New Area (Hebei), Handan 056038, China
[3]Key Laboratory of Intelligent Industrial Equipment Technology of Hebei Province, Handan 056038, China

Corresponding author: Xin-Jia Meng (mxjdoctor@126.com)

This work was supported by Hebei Natural Science Foundation under Grant E2021402052, and by Handan Science and Technology Research Program under Grant 21422021299.

ABSTRACT It is an important capacity to plan an optimal working path without collision for robotic arms. Rapid exploration random tree (RRT) is a popular algorithm in the field of path planning. However, the high computational cost caused by directly using RRT to plan the path of robotic arms hinders the practical application of RRT. In this paper, a variable sampling area RRT (VSA-RRT) algorithm is proposed to improve the computational efficiency of robotic arm path planning. An obstacle avoidance model based on limiting joint angle is first developed to judge the collision. Through this obstacle avoidance model, the collision judgment of robotic arm is converted into the joint angle judgment problem, and thus, the collision situation can be directly recognized with the given joint angles. Then, a sampling area dividing strategy is proposed on the basis of the limiting joint angles, where the motion space of robotic arm is divided into several subareas. Subsequently, VSA-RRT is proposed by integrating both the obstacle avoidance model and the sampling area dividing strategy. In VSA-RRT, the sampling area is automatically selected from the divided subareas, leading the sampling process to continuously move towards the target point, and the new sampling point is generated by using the obstacle avoidance model, ensuring that any new sampling point is collision-free. Finally, the effectiveness of the proposed VSA-RRT is demonstrated with three examples. The example results show that VSA-RRT decreases the sampling number by 44.98%, 19.95% and 15.00% compared with the basic RRT, RRT with SAD and APF-RRT respectively.

INDEX TERMS Variable sampling area, limiting joint angle, VSA-RRT, path planning, robotic arm.

I. INTRODUCTION

Benefiting from its excellent capability of task execution in dangerous, complex or inefficient context, robotic arms are widely used in electronics, vehicles, chemical and other industrial fields. It is an essential part of robotic arm application to plan an optimal working path with sufficient safety and reliability. This essential part is ordinarily completed with the aid of path planning technology. The technology of robotic arm path planning generally aims to find a feasible path that can transform the robotic arm from the initial configuration to the target configuration. To get the feasible path, the most direct and effective way is to use program-by-teaching model [1]. However, the use of teach pendant requires professional skills and knowledge, and it usually takes a considerable amount of time for planning a reasonable path [2]. Another different way is the simulation-based offline programming mode which is more efficient [3]. As a result of the development of flexible manufacturing and intelligence manufacturing, this offline

The associate editor coordinating the review of this manuscript and approving it for publication was P. Venkata Krishna.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

programming mode has become one focus in robot application and research [4].

In the offline programming mode, the path planning algorithm is used to find the feasible path. There are many well developed planning algorithms, while it is difficult to give a single criterion to classify all these algorithms in distinct categories [5]. For simple and clear understanding therefore, the common planning algorithms are surveyed chronologically. Dijkstra proposed his algorithm to obtain the shortest path with less data store [6], and later this algorithm was often adopted in optimal path planning of industrial end effector [7], [8]. Hart et al. [9] proposed a A* algorithm to find the minimum cost path between two points, and subsequently this algorithm continuingly was improved and successfully applied to robot path planning in various application scenarios [10], [11]. Khatib [12] proposed an artificial potential field (APF) method to plan a collision free path for robotic arms and mobile robots. Even if the original APF method has been proposed for several decades, the research about this method is still active. Some improvement and complement have been recently developed for APF method, such as membrane evolutionary APF [13], parallel evolutionary APF [14], and so on. Kavraki et al. [15] proposed probabilistic road map (PRM) algorithm to tackle the problems of robotic arm path planning in high dimensional configuration spaces. Stentz [16] proposed a D* algorithm to handle path cost optimization problem in which the cost parameters changed during the process of solution searching. LaValle and Kuffner [17] developed the rapid exploration random tree (RRT) algorithm to plan the collision-free kinodynamic path for robots. Besides, some bionic algorithms are also employed in robot path planning, for example, genetic algorithm [18], particle swarm optimization [19], ant colony optimization [20], and so on.

The RRT algorithm has been widely adopted in the fields of robot path planning, computer graphics and motion planning [21]. Although RRT shows excellent performance in many practical scenarios, its standard form has limitations in convergence speed, path quality and computational efficiency [22]. These challenges have led to the in-depth study of this algorithm. Karaman and Frazzoli [23] introduced the RRT* algorithm, which incorporates path reconnection and cost minimization strategies to improve path quality and ensure the asymptotic optimality. Jeong et al. [24] proposed a Q-RRT* algorithm by considering a set of vertices and their ancestors in the hypersphere, and as a result, the convergence rate is accelerated. Meng et al. [25] proposed a RRT* path planning approach based on exploring w-space to enhance the quality of the generated path for continuum arms. Zhang et al. [26] developed a BiRRT-APF algorithm to achieve the rapid convergence as well as reduce the computational burden for the path planning of robotic arm. Wang et al. [27] proposed an efficient motion planning algorithm by using bidirectional-unidirectional RRT extend function, which avoided the two-point boundary value problem. Yi et al. [28] proposed an improved P_RRT* algorithm for path planning of robotic arm, in which both potential function-based sampling method and twice expansion methods are adopted to improve the search efficiency. Ma et al. [29] proposed a probabilistic smoothing bidirectional RRT algorithm to improve the convergence speed. Cheng et al. [30] proposed an improved RRT-Connect path planning algorithm, in which the adaptive step strategy and the fixed sampling function are used to solve the low efficiency problem of RRT algorithm. Ding et al. [31] developed an extending path RRT* algorithm based on heuristic sampling in path expansion area, which can improve the node utilization and accelerate the convergence speed. Wang et al. [32] proposed a direction guidance RRT algorithm to balance the length of the path and the planning time. Feng et al. [33] proposed a directionally biased variable step APF-RRT* algorithm which use the directionally biased variable step sampling strategy and artificial potential field method to obtain the optimum path quality with efficient convergence rate.

The above improved RRT algorithms can enhance the performance of the path planning to a certain extent. However, in the path planning of robotic arm, the randomness of search range, the repeatability of collision judgment and the generation of large invalid nodes will lead to a significant computational cost. To overcome these problems, the obstacle avoidance model and the sampling area dividing strategy is introduced to the process of RRT algorithm in this paper. The main contributions are summarized as follows:

(1) The obstacle avoidance model is established through the limiting motion analysis of robotic arm relative to obstacles. Through the obstacle avoidance model, the collision-free configuration space of robotic arm can be directly obtained. Therefore, the collision judgment of robotic arm will be converted into the joint angle judgment problem, and the collision situation can be directly recognized with the given joint angles.

(2) A sampling area dividing strategy based on the limit joint angle is proposed to improve the sampling efficiency. Through this strategy, the sampling area is divided into several subareas. Each subarea restricts the different motion space of robot, and the motion range is gradually decreased towards the target position with the change of subarea.

(3) A variable sampling area RRT (VSA-RRT) algorithm is proposed by integrating both the obstacle avoidance model and the sampling area dividing strategy. The obstacle avoidance model is used to determine that the sampling is performed in collision-free area, which can ensure that any new sampling point is collision-free. The sampling area dividing strategy is adopted to guide the sampling process, in which the sampling area is automatically selected from the divided subareas according to the previous generated node, and thus, the newly generated node is ensured to move towards the target point.

The rest of this paper is organized as follows. In Section II, the background of the research is described. Section III gives a description of the obstacle avoidance model, sampling area dividing strategy and the VSA-RRT flowchart. In Section IV, three examples are used to demonstrate the validation of the proposed method. Section V concludes this work and makes suggestions for future work.

II. BACKGROUND

A. PATH PLANNING PROBLEM OF ROBOTIC ARM

The robot path planning is generally understood as a problem that aims to find a path between the starting point and target point. The path to be found usually possesses some significant features, such as shortest possible length and collision-free. For the path planning of robotic arm, the planned path often refers to the path that the end effector needs to pass through.

Since the position that the end-effector can reach is determined by the robot joint motion, the path planning of robotic arm can also be regarded as a problem that is to determine a series of configurations with the given starting configuration C_{init} and target configuration C_{goal}, where C_{init} and C_{goal} corresponds to the starting point and target point, respectively. To obtain the configurations corresponding to the planned path, it is better to resort to the forward kinematics and the inverse kinematics [34]. The inverse kinematics is used to compute the joint angles with a given path point, while the forward kinematics is used to compute the robot position with the given joint angles. In the path planning process, the collision at a given path point can be examined through inverse kinematics solution, forward kinematics solution and shortest distance calculation [19].

B. PATH PLANNING FOR ROBOTIC ARM BY RRT

The RRT algorithm belongs to sampling-based path planning method. For the path planning of robotic arm, RRT can be directly use to generate joint angle sampling point, where the joint angels corresponding to starting point and target point are given as initial condition, and then the path of end-effector can be obtained by forward kinematics solution [35], [36]. The pseudo-code of path planning for robotic arm by RRT is shown in Algorithm 1.

Algorithm 1 Path Planning for Robotic Arm by RRT

Input: $\boldsymbol{\theta}_{\text{init}}, \boldsymbol{\theta}_{\text{goal}}$;
Code 1: $\mathbf{V} \leftarrow \{\mathbf{V}, \boldsymbol{\theta}_{\text{init}}\}$, $\mathbf{E}=\emptyset$, $k = 0$
Code 2: **while** $k <= k_{\max}$ **do**
Code 3: $\boldsymbol{\theta}_{\text{rand}} \leftarrow$ Sample ($\boldsymbol{\theta}_{\min}, \boldsymbol{\theta}_{\max}$);
Code 4: $\boldsymbol{\theta}_{\text{near}} \leftarrow$ Nearest ($\mathbf{V}, \boldsymbol{\theta}_{\text{rand}}$);
Code 5: $\boldsymbol{\theta}_{\text{new}} \leftarrow$ Step ($\boldsymbol{\theta}_{\text{near}}, \boldsymbol{\theta}_{\text{rand}}$);
Code 6: if collision then
Code 7: return to Step 3;
Code 8: end if
Code 9: $\mathbf{V} \leftarrow \{\mathbf{V}, \boldsymbol{\theta}_{\text{new}}\}$;
Code10: $\mathbf{E} \leftarrow \{\mathbf{E}$, connect ($\boldsymbol{\theta}_{\text{near}}, \boldsymbol{\theta}_{\text{new}}$)};
Code 11: if reach ($\boldsymbol{\theta}_{\text{new}}, \boldsymbol{\theta}_{\text{goal}}$) then
Code 12: break;
Code 13: else
Code 14: $k \leftarrow k+1$;
Code 15: end if
Code 16: end while
Output:path

As shown in Algorithm 1, $\boldsymbol{\theta}_{\text{init}}$ and $\boldsymbol{\theta}_{\text{goal}}$ are the joint angle vector corresponding to C_{init} and C_{goal}, respectively. In the initial phase of the algorithm, the joint angle $\boldsymbol{\theta}_{\text{init}}$ is added into the node set $\mathbf{V}$, and the edge set $\mathbf{E}$ is set as empty. The initial iteration is set as $k = 0$. Then, a random joint angle $\boldsymbol{\theta}_{\text{rand}}$ is generated form the sampling area ($\boldsymbol{\theta}_{\min}, \boldsymbol{\theta}_{\max}$) (Code 3). In Code 4, the node $\boldsymbol{\theta}_{\text{near}}$ nearest to $\boldsymbol{\theta}_{\text{rand}}$ is selected from$\mathbf{V}$. The new joint angle $\boldsymbol{\theta}_{\text{new}}$ is generated with the specified step size in Code 5. Then, the collision judgement is performed (From Code 6 to Code 8). If collision, it returns to Code 3. Otherwise, Code 9 is performed. In Code 9, the new joint angle $\boldsymbol{\theta}_{\text{new}}$ is added into the node set. The edge connected from $\boldsymbol{\theta}_{\text{near}}$ to $\boldsymbol{\theta}_{\text{new}}$ is also added into edge set $\mathbf{E}$ (Code 10). The next step is to perform convergence judgment (From Code 11 to Code 15). If $\boldsymbol{\theta}_{\text{new}}$ reaches to $\boldsymbol{\theta}_{\text{goal}}$, the iteration will end, and the path planning is successful. Otherwise, the iteration is updated to $k = k + 1$. While $k <= k_{\max}$, the above steps will be repeated.

III. VARIABLE SAMPLING AREA RRT

The VSA-RRT algorithm integrates the obstacle avoidance model and the sampling area dividing strategy into the RRT algorithm to complete the path planning for robotic arm. In this section, the obstacle avoidance model is firstly established based on the limiting joint angle. Then, the sampling area dividing strategy is introduced. Lastly, the flowchart of VSA-RRT algorithm is described in detail.

A. OBSTACLE AVOIDANCE MODEL BASED ON LIMITING JOINT ANGLE

An obstacle avoidance strategy based on virtual body deformation has been recently developed by our team [37]. The main idea of this strategy is that there are limited positions for the robot to collide with obstacles when there are obstacles in the workspace of the robot. Thus, the limiting joint angles corresponding to these limiting positions can be computed to conduct collision judgment. Especially, instead of sphere envelope, the ellipsoidal envelope that has a high degree of adaptability is adopted to further improve the practicality of this strategy in this paper. Fig. 1 presents the principle of obstacle avoidance based on limiting joint angle.

In Fig. 1, θ_i ($i = 1, 2, 3, 4, 5, 6$) represents the ith joint angle, and L_i($i = 1, 2, 3, 4, 5, 6$) represents the length of the ith link. As shown in Fig. 1, once the relative positions of the robotic arm and obstacle are known, the limiting joint angles which indicate the limiting positions whether the links collide with obstacles can be determined. Since the obstacles are enveloped in ellipsoidal shapes, the limiting joint angle for each link can be obtained geometrically. Therefore, the

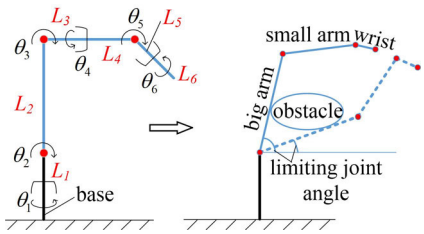

FIGURE 1. Basic principle of obstacle avoidance based on limiting joint angle.

obstacle avoidance models based on limiting joint angle are derived in the following part.

The limiting joint angle of the big arm involves joint angle θ_1 and joint angle θ_2. Therefore, the derivation of the obstacle avoidance model of big arm is conducted through θ_1 and θ_2. First, the big arm and the obstacle are enveloped into a cylinder and an ellipsoid, respectively, as shown in Fig.2. Based on these enveloped shapes, the big arm can be further simply represented as a straight line that is the axis of the cylinder, while the obstacle can be further represented as an expanded ellipsoid whose semi-axis length is increased by a cylinder radius r compared with its original length.

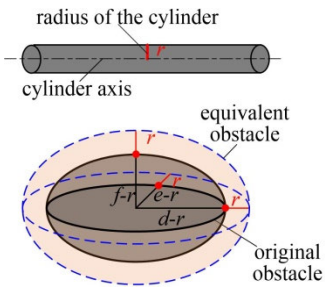

FIGURE 2. Enveloped shapes of big arm and obstacle.

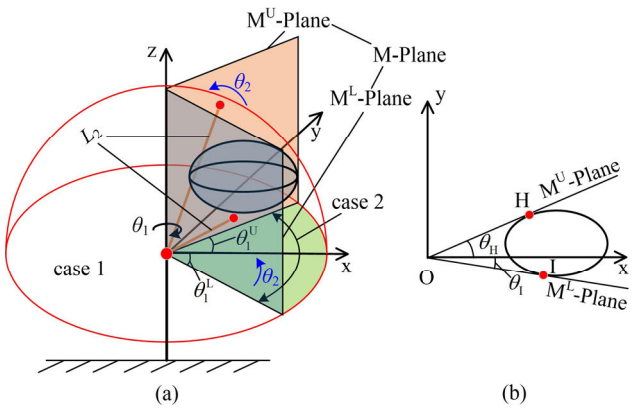

FIGURE 3. Schematic diagram of collision situation determined by θ_1.

Then, the coordinate system xyz-CSYS can be created as shown in Fig. 3a, where the centre of the second joint is taken as origin point, and the z-axis is along the rotation axis of θ_1. The equation of the obstacle in this coordinate system can be expressed as follows:

$$\frac{(x-a)^2}{d^2}+\frac{(y-b)^2}{e^2}+\frac{(z-c)^2}{f^2}=1 \tag{1}$$

where, (a, b, c) represents the coordinates of the centre of obstacle, and d, e and f represent the ellipsoid semi-axis corresponding to x-axis, y-axis and z-axis, respectively.

Fig.3a shows the collision situations that can be determined by θ_1, which are classified into two cases. For case 1, the obstacle is not within the scope of entity motion space of big arm, and thus, the collision will never occur. For case 2, although the obstacle is already within the scope of entity motion space of big arm, it is still not sure that the collision is certain to occur, which should be further analyzed with θ_2 according to the motion law of robotic arm.

Case 1 and case 2 can be recognized by the limiting angle of θ_1, which has two limiting values, the maximum limiting value θ_1^{U} and minimum limiting value θ_1^{L}. In fact, θ_1^{U} and θ_1^{L} correspond to the situations that M plane is tangent to the obstacle (see $\mathrm{M^U}$-plane and $\mathrm{M^L}$-plane in Fig.3a). In order to conveniently derive the formula of θ_1^{U} and θ_1^{L}, the obstacle and M plane are projected on x-o-y plane, as shown in Fig 3b. The formulas of θ_{I} and θ_{H} corresponding to the points of tangency are given by:

$$\begin{aligned}
&\theta_i(i=\mathrm{I}\text{ or }\mathrm{H})\\
&\quad=(1)90,\text{ when } x_i=0, y_i>0;\\
&\quad\ (2)-90,\text{ when } x_i=0, y_i<0;\\
&\quad\ (3)\arctan(y_i/x_i),\text{ when } x_i>0;\\
&\quad\ (4)\pi+\arctan(y_i/x_i),\text{ when } x_i=0,\ y_i\geq 0;\\
&\quad\ (5)-\pi+\arctan(y_i/x_i),\text{ when } x_i=0,\ y_i<0\\
&\quad\ \text{then } \theta_1^{\mathrm{U}}=\max(\theta_{\mathrm{I}},\theta_{\mathrm{H}}),\ \theta_1^{\mathrm{L}}=\min(\theta_{\mathrm{I}},\theta_{\mathrm{H}})
\end{aligned} \tag{2}$$

where $(x_i, y_i, 0)$ (for $i=\mathrm{I}$ or H) is the coordinates of point I or point H in xyz-CSYS. The values of coordinates of point I and point H can be obtained through the following equation:

$$\begin{cases} y=x\tan\theta_1\\ \dfrac{(x-a)^2}{d^2}+\dfrac{(y-b)^2}{e^2}=1\\ (4d^2-4a^2)t^2+8abt-4b^2+4e^2==0 \end{cases} \tag{3}$$

After obtaining the θ_{I} and θ_{H}, θ_1^{U} and θ_1^{L} are given by:

$$\begin{cases} \theta_1^{U}=\max(\theta_{\mathrm{I}},\theta_{\mathrm{H}})\\ \theta_1^{L}=\min(\theta_{\mathrm{I}},\theta_{\mathrm{H}}) \end{cases} \tag{4}$$

Consequently case 1 and case 2 can be summarized as follows:

$$\begin{cases} \text{case 1: } \theta_1\notin(\theta_1^{\mathrm{L}},\theta_1^{\mathrm{U}})\\ \text{case 2: } \theta_1\in(\theta_1^{\mathrm{L}},\theta_1^{\mathrm{U}}) \end{cases} \tag{5}$$

In case 2, both θ_1 and θ_2 need to be used to analyze the collision. For a fixed θ_1 belonging to case 2, whether a collision will occur needs to be determined through θ_2, which also

need to be classified into several different situations. These different situations can be recognized by the judgment that whether the big arm intersects with the intersecting ellipse (formed by the intersection of M plane and the obstacle, as shown in Fig 4).

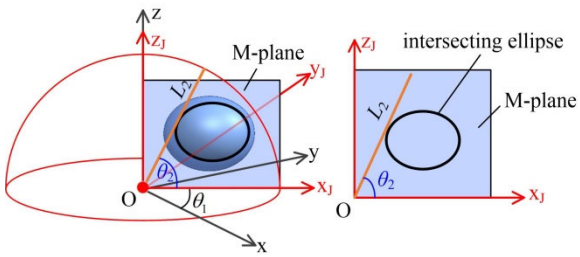

FIGURE 4. Intersecting ellipse of M plane and the obstacle.

In Figure 4, a new coordinate system $x_Jy_Jz_J$-CSYS is created by taking the original z-axis of xyz-CSYS as z_J-axis, and the x_J-axis and y_J-axis are established by rotating xoy around z-axis by θ_1. In fact, the x_Joz_J is just the M plane, that is, the motion plane of big arm driven by θ_2. The intersecting ellipse is the intersection line of the M plane and the obstacle. The equation of the intersecting ellipse therefore, can be represented as:

$$\begin{cases} \dfrac{(x_J\cos\theta_1 - a)^2}{d^2} + \dfrac{(x_J\sin\theta_1 - b)^2}{e^2} + \dfrac{(z_J - c)^2}{f^2} = 1 \\ y_J = 0 \end{cases} \tag{6}$$

where, (x_J, y_J, z_J) is the coordinate in $x_Jy_Jz_J$-CSYS.

Based on the equation of the intersecting ellipse and the length of big arm, the collision situations determined by θ_2 are classified into four cases, which are shown in Fig.5. It should be noted that Fig.5 shows the entity motion of the big arm in the x_Joz_J plane, which is driven by θ_2.

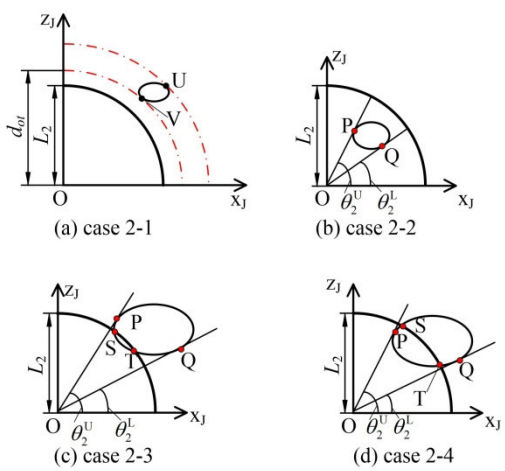

FIGURE 5. Schematic diagram of collision situation determined by θ_2.

In Fig. 5, OP and OQ are the tangent lines about the big arm and intersection line, and d_{o1} and d_{o2} respectively represent the maximum distance and minimum distance between coordinate origin and intersecting ellipse. S and T are the intersection points of the intersecting ellipse and the circle with the length of big arm as the radius (named as big arm circle for short, as shown in Fig.5). According to position relationship between the big arm and obstacle, the judgment of these four cases (respectively named as case 2-1, case2-2, case2-3 and case2-4) is given as follows:

$$\begin{cases} \text{case 2-1: } d_{ot} > L_2 \\ \text{case 2-2: } m_2 < L_2 \\ \text{case 2-3: } d_{ot} < L_2 \,\&\, n_2 > L_2 \\ \text{case 2-4: } n_2 < L_2 \,\&\, m_2 > L_2 \end{cases} \tag{7}$$

where

$$\begin{cases} m_2 = \max(d_{OP}, d_{OQ}) \\ n_2 = \min(d_{OP}, d_{OQ}) \\ d_{ot} = \min(d_{o1}, d_{o2}) \\ d_{o1} = \sqrt{x_U^2 + z_U^2},\ d_{o2} = \sqrt{x_V^2 + z_V^2} \\ d_{OP} = \sqrt{x_P^2 + z_P^2}, d_{OQ} = \sqrt{x_Q^2 + z_Q^2} \end{cases} \tag{8}$$

where d_{OP} and d_{OQ} are the length of the OP and OQ, respectively. $(x_U, 0, z_U)$ and $(x_V, 0, z_V)$ are the coordinates of points U and V, respectively, and they can be obtained by:

$$\begin{cases} \dfrac{(x_J\cos\theta_1 - a)^2}{d^2} + \dfrac{(x_J\sin\theta_1 - b)^2}{e^2} + \dfrac{(z_J - c)^2}{f^2} = 1 \\ \dfrac{x_J(e^2\cos^2\theta_1 + d^2\sin^2\theta_1)}{ae^2\cos\theta_1 + bd^2\sin\theta_1} = \dfrac{z_J}{c} \end{cases} \tag{9}$$

$(x_P, 0, z_P)$ and $(x_Q, 0, z_Q)$ are the coordinate of point P and point Q, respectively, and they can be obtained by:

$$\begin{cases} \dfrac{(x_J\cos\theta_1 - a)^2}{d^2} + \dfrac{(x_J\sin\theta_1 - b)^2}{e^2} + \dfrac{(z_J - c)^2}{f^2} = 1 \\ z_J = x_J\tan\theta_2 \\ b_{PQ}^2 - 4a_{PQ}c_{PQ} = 0 \end{cases} \tag{10}$$

where

$$\begin{cases} a_{PQ} = e^2f^2\cos\theta_1 + d^2f^2\sin\theta_1 + d^2e^2\tan\theta_2 \\ b_{PQ} = 2ae^2f^2\cos\theta_1 + 2bd^2f^2\sin\theta_1 + 2cd^2e^2\tan\theta_1 \\ c_{PQ} = ae^2f^2 + bd^2f^2 + cd^2e^2 - d^2e^2f^2 \end{cases} \tag{11}$$

For the case 2-1, as shown in Fig. 5(a), the length of big arm L_2 is less than d_{ot}, that is, the big arm circle does not intersect with the intersecting ellipse, indicating the big arm will never collide with obstacle. Therefore, in this case, the limiting value of θ_2 does not need to be taken into account.

For the case 2-2, as shown in Fig. 5(b), the length of big arm L_2 is bigger than m_2, that is, the big arm circle and the intersecting ellipse have the possibility of intersection. Therefore, the limiting values θ_2^L and θ_2^U should be solved,

and the collision will not occur as long as θ_2 does not belong to $(\theta_2^{\mathrm{L}}, \theta_2^{\mathrm{U}})$. θ_2^{L} and θ_2^{U} can be obtained by the following equation:

$$\begin{aligned}
&\theta_i(i = \mathrm{P} \text{ or } \mathrm{Q}) \\
&\quad = (1)90, \text{ when } x_i = 0, z_i > 0; \\
&\qquad (2) - 90, \text{ when } x_i = 0, z_i < 0; \\
&\qquad (3)\arctan(z_i/x_i), \text{ when } x_i > 0; \\
&\qquad (4)\pi + \arctan(z_i/x_i), \text{ when } x_i = 0,\ z_i \geq 0; \\
&\qquad (5) - \pi + \arctan(z_i/x_i), \text{ when } x_i = 0,\ z_i < 0 \\
&\qquad \text{then } \theta_2^{\mathrm{U}} = \max(\theta_{\mathrm{P}}, \theta_{\mathrm{Q}}), \theta_2^{\mathrm{L}} = \min(\theta_{\mathrm{P}}, \theta_{\mathrm{Q}})
\end{aligned} \tag{12}$$

where $(x_i, y_i, 0)$ (for $i = \mathrm{P}$ or Q) is the coordinates of point P or point Q in $x_J y_J z_J$-CSYS, and these values can be obtained by solving equation (10).

For the case 2-3, as shown in Fig. 5(c), the length of big arm L_2 is bigger than d_{ot} and is less than n_2, indicating the big arm circle and the intersecting ellipse also have the possibility of intersection. In this case the $\theta L2$ and $\theta U2$ can be obtained by the following equation:

$$\begin{aligned}
&\theta_i(i = \mathrm{S} \text{ or } \mathrm{T}) \\
&\quad = (1)90, \text{ when } x_i = 0, z_i > 0; \\
&\qquad (2) - 90, \text{ when } x_i = 0, z_i < 0; \\
&\qquad (3)\arctan(z_i/x_i), \text{ when } x_i > 0; \\
&\qquad (4)\pi + \arctan(z_i/x_i), \text{ when } x_i = 0,\ z_i \geq 0; \\
&\qquad (5) - \pi + \arctan(z_i/x_i), \text{ when} x_i = 0,\ z_i < 0 \\
&\qquad \text{then } \theta_2^{\mathrm{U}} = \max(\theta_{\mathrm{S}}, \theta_{\mathrm{T}}), \theta_2^{\mathrm{L}} = \min(\theta_{\mathrm{S}}, \theta_{\mathrm{T}})
\end{aligned} \tag{13}$$

where $(x_i, y_i, 0)$ (for $i = \mathrm{S}$ or T) is the coordinates of point S or point T in $x_J y_J z_J$-CSYS, respectively, and they can be obtained by solving the following equation:

$$\begin{cases} \dfrac{(x_J \cos\theta_1 - a)^2}{d^2} + \dfrac{(x_J \sin\theta_1 - b)^2}{e^2} + \dfrac{(z_J - c)^2}{f^2} = 1 \\ z_J^2 + x_J^2 = L_2^2 \end{cases} \tag{14}$$

For the case 2-4, as shown in Fig. 5(d), the length of big arm L_2 is bigger than n_2 and is less than m_2. The potential collision is also existed between the big arm and the obstacle in this case, and therefore, the θ_2^L and θ_2^U can be obtained by the following equation:

$$\begin{cases} \text{if OP>OQ}, \theta_{\text{mn-1}} = \theta_{\mathrm{Q}}; \\ \text{if OQ>OP}, \theta_{\text{mn-1}} = \theta_{\mathrm{P}}; \\ \text{if } |\theta_{\mathrm{S}} - \theta_{\text{mn-1}}| > |\theta_{\mathrm{T}} - \theta_{\text{mn-1}}|, \theta_{\text{mn-2}} = \theta_{\mathrm{S}}; \\ \text{if } |\theta_{\mathrm{S}} - \theta_{\text{mn-1}}| < |\theta_T - \theta_{\text{mn-1}}|, \theta_{\text{mn-2}} = \theta_{\mathrm{T}}. \end{cases}$$
$$\text{then } \theta_2^{\mathrm{U}} = \max(\theta_{\text{mn-1}}, \theta_{\text{mn-2}}), \theta_2^{\mathrm{L}} = \min(\theta_{\text{mn-1}}, \theta_{\text{mn-2}}) \tag{15}$$

Through the above analysis, the obstacle avoidance model based on limiting joint angle for big arm can be summarized as follows:

$$\begin{cases} \textit{if}\ \ \theta_1 \in (\theta_1^{\mathrm{L}}, \theta_1^{\mathrm{U}}) \& L_2 > d_{ot} \& \theta_2 \in (\theta_2^L, \theta_2^U), \\ \quad \text{Collision}; \\ \text{Otherwise, not Collision.} \end{cases} \tag{16}$$

When the joint angles of robotic arm are known, weather the robotic arms collide with obstacles can be directly judged by putting the known joint angles into the obstacle avoidance model based on limiting joint angle.

Considering its own structural characteristics, only the limiting angles of the big arm link have been derived. This is because other limiting angles corresponding to the rest links can be easily obtained through the coordinate transformation, which can refer to Ref. [37].

B. SAMPLING AREA DIVIDING STRATEGY

In order to quickly reach to the target configuration from the starting configuration, the sampling area dividing strategy is proposed on the basis of the limiting joint angle. Among all joint angles, the first joint angle (that is θ_1) has a significant impact on the collision of robotic arm. Therefore, the sampling area dividing is only applicable to the first joint angle. For other joint angles, the sampling area does not need to be divided, but the collision-free area is directly adopted as the sampling area. The schematic diagram of sampling area dividing is shown in Fig. 6.

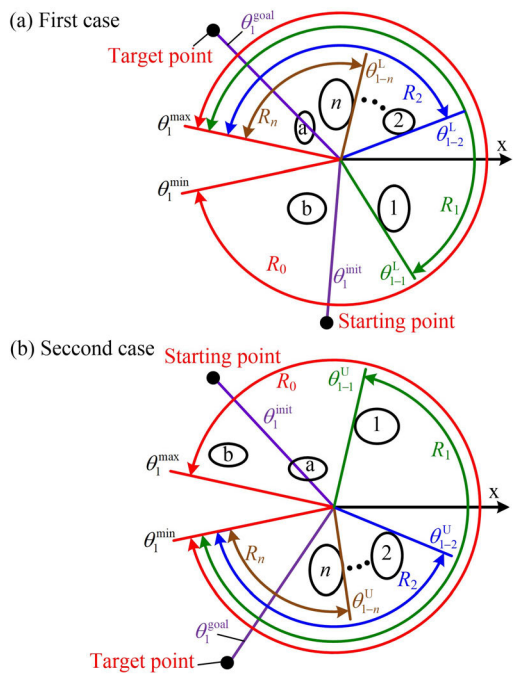

FIGURE 6. Schematic diagram of sampling area dividing.

As shown in Fig. 6, the sampling area dividing includes two cases. One case is that the first joint angle of starting configuration is less than that of target configuration, that is, $\theta_1^{\text{init}} < \theta_1^{\text{goal}}$. Another case is that the first joint angle

of starting configuration is larger than that of target configuration, that is, $\theta_1^{\text{init}} > \theta_1^{\text{goal}}$. The process of sampling area dividing mainly concludes two steps, which can be described as follows.

Step 1: Obstacle numbering

When numbering, it only needs to number the obstacles which completely located within the range from the first joint angle of starting configuration to first joint angle of target configuration. The obstacles a and b do not completely locate within the range from θ_1^{init} to θ_1^{goal}. Therefore, these two obstacles will not be numbered. The number sequence is that the obstacle nearest to the initial joint angle is numbered to 1. Then, the numbering is completed in sequence, as shown the obstacle number from 1 to n in Fig. 6, where n is the number of obstacles numbered.

Step 2: Area dividing

By area dividing, the sampling area of the first joint angle is divided into several subareas. The number of sampling areas is determined by the number of numbered obstacles. The sampling area can be given by:

$$\begin{cases} R_i = \left(\theta_1^{\min}, \theta_1^{\max}\right), \ i = 0; \\ R_i = (\theta_{1-i}^{\text{L}}, \theta_1^{\max}), \text{ when } \theta_1^{\text{init}} < \theta_1^{\text{goal}}, \\ \qquad\qquad \text{for } i = (1, 2, 3, \ldots, n); \\ R_i = (\theta_1^{\min}, \theta_{1-i}^{\text{U}}), \text{ when } \theta_1^{\text{init}} > \theta_1^{\text{goal}}, \\ \qquad\qquad \text{for } i = (1, 2, 3, \ldots, n). \end{cases} \tag{17}$$

where R_i stands for the ith sampling area. $\theta_1^{\min}$ and $\theta_1^{\max}$ are respectively the allowable minimum and maximum of θ_1 of robot. θ_{1-i}^{L} and θ_{1-i}^{U} are respectively the limiting value relative to the ith obstacle, which can be obtained by the established obstacle avoidance model.

The dividing strategy indicates that the range from R_0 to R_{n} is gradually decreased towards the target position, which can ensure that the sampling process continuously moves towards the target point.

C. FLOWCHART OF VSA-RRT

The flowchart of VSA-RRT is shown in Fig. 7. The specific steps of the proposed VSA-RRT are described as follows.

Step 1: Initialization

The initial cycle number is set as $k = 0$, and the initial cycle number for the step of selecting sampling area (Step 6) is set as $\mu = 0$. The other relative parameters are defined by:

$$\begin{cases} \mathbf{P}_{start} = (x_{start}, y_{start}, z_{start}) \\ \mathbf{P}_{\text{goal}} = (x_{start}, y_{start}, z_{start}) \\ \boldsymbol{\theta}_{init} = (\theta_1^{\text{init}}, \theta_2^{\text{init}}, \ldots, \theta_t^{\text{init}}) \\ \boldsymbol{\theta}_{goal} = (\theta_1^{\text{goal}}, \theta_2^{\text{goal}}, \ldots, \theta_t^{\text{goal}}) \\ d = (d_1, d_2, \ldots, d_t) \\ \mathbf{V}_{\text{angle}} = [\boldsymbol{\theta}_{init}] \\ \mathbf{V}_{\text{point}} = [\mathbf{P}_{start}] \\ \mathbf{T} = [\mathbf{V}_{\text{angle}}, \mathbf{V}_{\text{point}}] \end{cases} \tag{18}$$

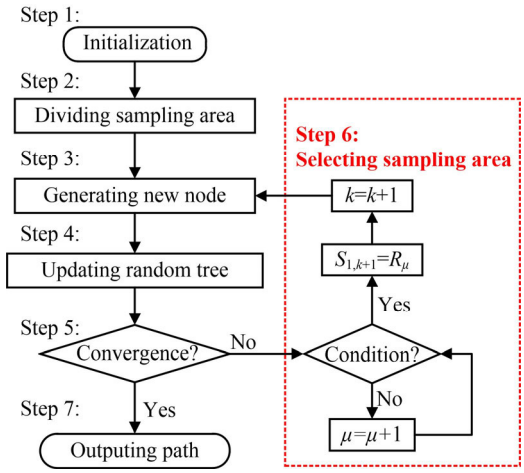

FIGURE 7. Flowchart of VSA-RRT.

where, $\mathbf{P}_{\text{start}}$ and $\mathbf{P}_{\text{goal}}$ are the coordinate of starting point and target point, respectively. $\theta_j^{\text{init}}(j=1, 2, \ldots, t)$ is the jth joint angle of starting configuration corresponding to starting point, and θ_j^{goal} is the jth joint angle of target configuration corresponding to target point. $\mathbf{d}$ is the vector of step size, and d_j is the step size of jth joint angle. t is the number of joint angles. $\mathbf{V}_{\text{angle}}$ is the joint angle node set. $\mathbf{V}_{\text{point}}$ is the path point set. $\mathbf{T}$ is the random tree, which is the set of joint angle node and path point node.

Step 2: Dividing sampling area

The sampling area is divided according to the sampling area dividing strategy. The sampling area set is denoted as $\mathbf{R}=\{R_0, R_1, \ldots, R_j, \ldots, R_{\text{n}}\}$. Moreover, the initial sampling area of θ_1 is defined as R_0, that is, $S_{1k} = R_0$ when $k = 0$, where S_{1k} is the sampling area of θ_1 selected in k^{th} iteration.

Step 3: Generating new node

In this step, the new nodes for each joint angle will be generated. This step can be mainly described the following seven steps.

Step 3-1: Initialization of Step 3

The initial cycle number is set as $j = 1$. The sampling area of θ_1 is set as S_{1k}. the initial new node of joint angle is set as $\boldsymbol{\theta}_k^{new} = \emptyset$, where symbol 'Ø' stands for 'empty set'.

Step 3-2: Generating sampling point for θ_j

The sampling point for θ_j is randomly generated in the sampling area S_{jk}, which is defined by:

$$\theta_{jk}^{\text{rand}} = S_{jk}^L + (S_{jk}^U - S_{jk}^L) rand(1) \tag{19}$$

where, $\theta_{jk}^{\text{rand}}$ is the random node for θ_j generated in k^{th} iteration. S_{jk}^{L} and S_{jk}^{U} are the minimum and maximum value of sampling area S_{jk} (S_{jk} is the sampling area of θ_j in k^{th} iteration), respectively. rand(1) is a random number within 0 to 1.

Step 3-3: Selecting nearest node for θ_j

The nearest node $\theta_{jk}^{\text{nearest}}$ for θ_j is selected in random tree, which can be given by:

$$\theta_{jk}^{\text{nearest}} = \begin{cases} \theta_j^{\text{init}}, for\ j = 1, k = 0; \\ \theta_{ji} = \text{find}(\min\limits_{i=1}^{m_1} \left|\theta_{jk}^{rand} - \theta_{ji}\right|), \\ \qquad for\ j = 1, k \neq 0, \theta_{ji} \in \mathbf{T}; \\ \theta_{ji} = \text{find}(\min\limits_{i=1}^{m_2} \left|\theta_{jk}^{rand} - \theta_{ji}\right|), \\ \qquad for\ j \neq 1, k \neq 0, \theta_{ji} \in \mathbf{T}, \\ \qquad S_{jk}^{\text{L}} < \theta_{ji} < S_{jk}^{\text{U}}. \end{cases} \tag{20}$$

where, $\theta_{jk}^{\text{nearest}}$ is the nearest node of θ_j in k^{th} iteration. m_1 is the number of θ_1 in $\mathbf{T}$. m_2 is the number of θ_j which satisfies the condition of $S_{jk}^{\text{L}} < \theta_j < S_{jk}^{\text{U}}$ in $\mathbf{T}$. 'find(Ψ)' represents searching the θ_{ji} which satisfies the condition of Ψ.

Step 3-4: Generating new node for θ_j

The new node is generated by step size, which can be calculated by:

$$\theta_{jk}^{\text{new}} = \left(\theta_{jk}^{\text{rand}} - \theta_{jk}^{\text{nearest}}\right) d_j + \theta_{jk}^{\text{nearest}} \tag{21}$$

where, θ_{jk}^{new} is the new node of θ_j generated in k^{th} iteration.

Step 3-5: Updating parameters

The relative parameters need to be updated in this step, which can be given by:

$$\begin{cases} \text{First update: } \boldsymbol{\theta}_k^{\text{new}} = [\boldsymbol{\theta}_k^{\text{new}}, \theta_{jk}^{\text{new}}] \\ \text{Second update: } j = j + 1 \end{cases} \tag{22}$$

Step 3-6: Judgement

When the cycle number is larger than the maximum number of joint angle (that is, $j > t$), the new node $\boldsymbol{\theta}_k^{\text{new}}$ will be outputted to perform Step 4. Otherwise, the Step 3-7 will be performed.

Step 3-7: Updating sampling area

The sampling area S_{jk} is updated according to the collision-free region of θ_j which is calculated by the obstacle avoidance model. When the collision-free region of θ_j is composed of multiple discontinuous intervals, the new sampling area will be random selected from the collision-free region, and then, the Step 3-2 will continue to be performed.

Step 4: Updating random tree

Firstly, the new joint angle node generated by Step 3 is added into the joint angle note set. Then, the new path point is added into the path point set. Lastly, the random tree is updated. These updating can be given by:

$$\begin{cases} \mathbf{V}_{\text{angle}} = [\mathbf{V}_{\text{angle}}; \boldsymbol{\theta}_k^{\text{new}}] \\ \mathbf{V}_{\text{point}} = [\mathbf{V}_{\text{point}}; \mathbf{P}_k^{\text{new}}] \\ \mathbf{T} = [\mathbf{V}_{\text{angle}}, \mathbf{V}_{\text{point}}] \end{cases} \tag{23}$$

where, $\mathbf{P}_k^{\text{new}}$ is the path point coordinate corresponding to the new joint angle node $\boldsymbol{\theta}_k^{\text{new}}$, which can be obtained by the forward kinematics solution.

Step 5: Convergence judgement

The convergence criterion is set as follows:

$$\begin{cases} k > k_{\max} \\ or \\ \left|\theta_{jk}^{new} - \theta_j^{goal}\right| \leq \varepsilon_j, j = 1, 2, \ldots, t \end{cases} \tag{24}$$

where, ε_j is the convergence precision corresponding to the jth joint angle. $k_{\max}$ is the allowable maximum iterations.

When the convergence criterion is satisfied, the step 7 will be performed. Otherwise, the step 6 will be performed.

Step 6: Selecting sampling area

This step is mainly used to select the sampling area for θ_1 of next iteration. Firstly, the area where the new node of θ_1 obtained by the current iteration located is judged by the condition shown in Eq. (25).

$$\begin{cases} \theta_{1k}^{\text{new}} > R_{\mu+1}^{\text{U}}, \text{for } \theta_1^{\text{init}} > \theta_1^{\text{goal}} \\ \theta_{1k}^{\text{new}} < R_{\mu+1}^{\text{L}}, \text{for } \theta_1^{\text{init}} \leq \theta_1^{\text{goal}} \end{cases} \tag{25}$$

where, $R_{\mu+1}^{\text{L}}$ and $R_{\mu+1}^{\text{U}}$ are the lower limit and upper limit of area $R_{\mu+1}$ which is divided by step 2. When the condition is not satisfied, the cycle number is updated to $\mu = \mu + 1$, and then the condition is continued to be judged. When the condition is satisfied, the area R_μ is selected as the sampling area of θ_1 in next iteration (that is, $S_{1,k+1} = R_\mu$). Then, the overall iteration is updated to $k = k + 1$, and meanwhile the step 3 will continue to be performed.

Step 7: Outputting path

The path is obtained by searching from the random tree according to the principle of minimum distance. The path searching is started from the target point. In the random tree, the node nearest to the target point is searched as the next path point, which is denoted as P1. Then, the node nearest to the point P1 is continued to search. The searching is performed in sequence until the nearest node obtained is the starting point. Finally, the path points searched are outputted as the planned path, and meanwhile the joint nodes corresponding to the path points is also outputted as the joint configuration of robot.

IV. EXAMPLE VALIDATION

In this section, three examples are designed to validate the proposed method. The ABBIRB120 robot is used for the validation. The established coordinate system of ABBIRB120 robot is shown in Fig. 8, while the D-H parameters and the motion limit of each joint angle are listed in Table 1.

A. VERIFICATION OF OBSTACLE AVOIDANCE MODEL

The first example is mainly used to verify the effectiveness of the established obstacle avoidance model. Two ellipsoid obstacles are designed to complete the validation. The obstacle parameters are listed in Table 2. MATLAB is used to complete the simulation. The results are listed in Table 3 and Fig. 9.

From Table 3, the limit ranges of θ_1 are respectively [−9.77°, 56.41°] and [13.83°, 53.55°] for obstacle A and B. As shown in Fig. 9(a.1) and Fig. 9 (b.1), the obstacle A and B

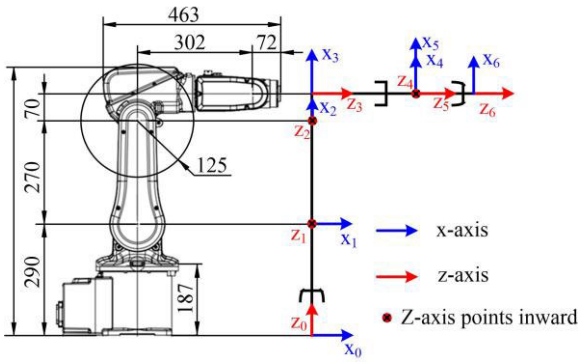

FIGURE 8. Coordinate system of ABBIRB120 robot.

TABLE 1. D-H Parameters and motion limit Of ABBIrb120 robot.

link	θ_i/rad	d/mm	a/mm	alpha/rad	Motion limit
1	θ_1	290	0	−pi/2	−165°~+165°
2	θ_2	0	270	0	−110°~+110°
3	θ_3	0	70	−pi/2	−110°~+70°
4	θ_4	302	0	pi/2	−160°~+160°
5	θ_5	0	0	−pi/2	−120°~+120°
6	θ_6	72	0	0	−400°~+400°

TABLE 2. Obstacle parameters.

Obstacle	(a, b, c)	d	e	f
A	(30,15,25)	15	20	10
B	(20,15,10)	5	10	15

are completely within the range of [−9.77°, 56.41°] and [13.83°, 53.55°], respectively. This demonstrates that when θ_1 does not belong to the limit ranges, the big arm does not collide with obstacle no matter what value θ_2 is. Otherwise, when θ_1 belongs to the limit ranges, it is uncertain whether the big arm collides with obstacle, which need to perform the collision assessment by θ_2. For each obstacle, three values of θ_1 are used to test the collision. For obstacle A, when θ_1 is taken as 0°, 30° and 55°, the collision areas of θ_2 shown from Fig. 9(a.2) to Fig. 9(a.4), are [31.69°, 52.92°], [25.68°, 56.17°], and empty set, respectively. Fig. 9(a.2), Fig. 9(a.3) and Fig. 9(a.4) are corresponding to the case 2-3, case 2-4, and case 2-1 shown in Fig. 5, respectively. For obstacle B, when θ_1 is taken as 30°, 45°, and 60°, the collision areas of θ_2 shown from Fig. 9(b.2) to Fig. 9(b.4) are [−10.04°, 46.86°], [−7.29°, 42.20°], and empty set, respectively. Fig. 9(b.2) and Fig. 9(b.3) are both corresponding to the case 2-2 shown in Fig. 5, while Fig. 9(b.4) is corresponding to the case 1 shown in Fig. 3. As shown in Fig. 9, the big arm does not collide with obstacle A and B outside the collision area of θ_2. These indicate that the collision-free area can be directly given by the established obstacle avoidance model, and the established obstacle avoidance model can be effectively used to guide the path planning of obstacle avoidance.

B. VERIFICATION OF VSA-RRT

The second example is primarily designed to validate the effectiveness of the proposed VSA-RRT. Two obstacles are also designed to complete this validation. The relevant parameters are listed in Table 4, and the simulation results are shown in Fig. 10.

In Fig. 10, invalid node is the generated node which collide with the obstacles. Fig. 11(a) shows the results obtained by the basic RRT method described in Part A of Section II. Fig. 11(c) shows the results obtained by the proposed method. In addition, to further explore the effect of sampling area dividing strategy on the performance of the algorithm, a RRT with sampling area dividing (RRT with SAD) algorithm is also designed. The RRT with SAD only adopted the sampling area dividing strategy of the first joint angle, but not considering the collision impact of other joint angles, when sampling. The results by RRT with SAD is shown in Fig. 11(b).

As shown in Fig. 10, for the sampling area of θ_1, the basic RRT method haves only one area R_0, while the RRT with SAD method and the VSA-RRT algorithm have three areas of R_0, R_1, and R_2. R_0 is the range from −165° to +165° which is the motion limit of the first joint angle of robot shown in Table 1. R_1 is the range from −165° to 20.3002°, and R_2 is the range from −165° to −16.1536°. The maximum values of R_1 and R_2 (that is, 20.3002° and −16.1536°) are respectively the maximum limiting angle corresponding to obstacle 1 and obstacle 2, which is calculated by the obstacle avoidance model. The sampling area of other joint angle for basic RRT method and RRT with SAD method are both the motion limit range, which is a fixed area. However, the sampling area of other joint angle for VSA-RRT algorithm needs to be calculated by using obstacle avoidance model, which is not a fixed area but changes with the values of the joint angle in its front.

Moreover, as shown in Fig. 10, the number of generated invalid nodes are 928, 690, and 0, respectively, for three different methods. This is because that, the basic RRT method neither adopts the sampling area dividing strategy nor considering the collision impact on sampling area. For the basic RRT, the sampling area is the motion limit area listed in table 1. Therefore, the number of invalid nodes generated by basic RRT method is the largest one. For RRT with SAD, it adopts the sampling area strategy. The number of obstacles included in the sampling area from R_0 to R_n is gradually decreased. When the sampling point located in the next area, the previous sampling area will no longer be used in the subsequent sampling process. Consequently, when sampling for the first joint angle, the number of nodes sampled in the area which may cause the collision will be decreased. However, the RRT with SAD method does not consider the collision impact of other joint angles, and thereby the number of invalid nodes may be still generated but is less than that by the basic RRT. For the proposed VSA-RRT, the sampling area dividing strategy and the collision impact is simultaneously considered. During sampling, the collision area was completely avoided. Therefore, the proposed VSA-RRT algorithm will not generate invalid nodes. The invalid node is not adopted as the node on the planned path, which need to be abandoned. The more invalid nodes there are, the longer time will

TABLE 3. Calculation results Of θ_2.

Obstacle	$[\theta_1^L, \theta_1^U]$	No.	θ_1	L_2	m_2	n_2	d_{ot}	θ_2^L	θ_2^U	Collision area
A	[–9.77°, 56.41°]	1	0	35	41.4	35.1	30.9	31.69°	52.92°	[31.69°, 52.92°]
		2	30°	35	45.4	34.7	29.4	25.68°	56.17°	[25.68°, 56.17°]
		3	55°	35	44.5	39.7	38.1	/	/	Ø
B	[13.83°, 53.55°]	1	30°	35	31.7	23.6	20.0	–10.04°	46.86°	[–10.04°, 46.86°]
		2	45°	35	33.5	26.8	22.7	–7.29°	42.20°	[–7.29°, 42.20°]
		3	60°	35	/	/	/	/	/	Ø

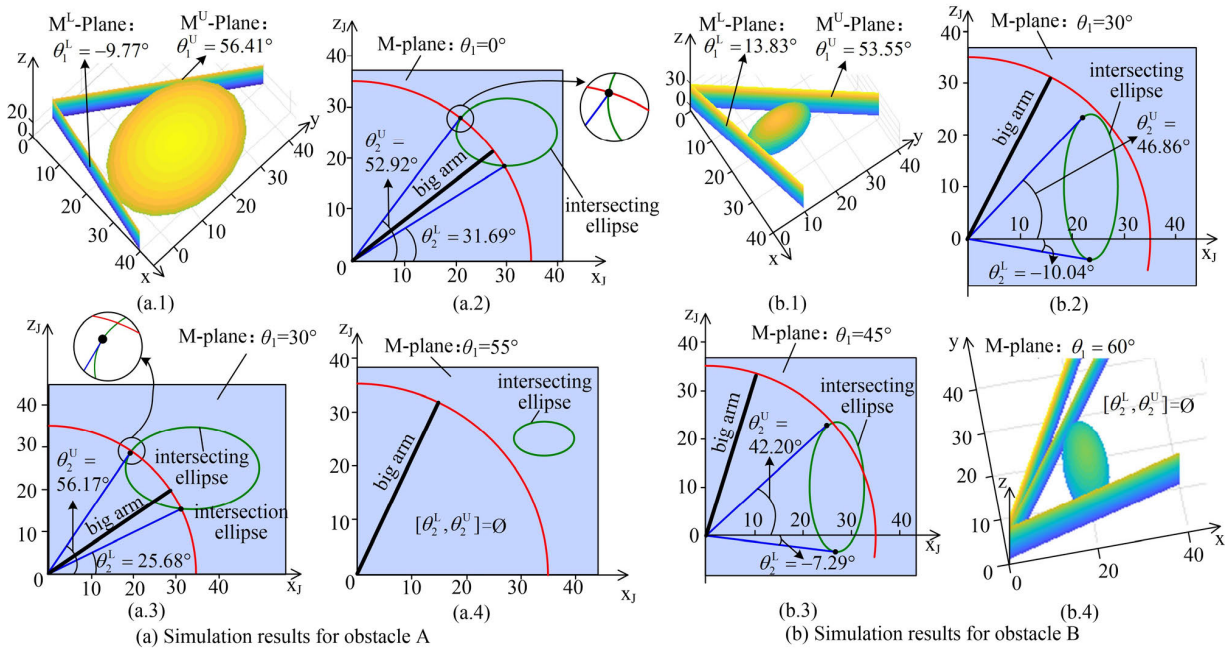

FIGURE 9. Simulation results of first example.

TABLE 4. Relative parameters of second example.

Object	Parameter	Value
Robot	Length of big arm /mm	340
	Length of small arm /mm	302
	Length of Wrist /mm	72
	Envelope radius of big arm /mm	100
	Envelope radius of small arm /mm	70
	Envelope radius of wrist /mm	60
	Joint angle of starting configuration /°	[55.4077, –29.4159, –28.5964, 0, 58.0123, –124.5923]
	Joint angle of target configuration /°	[–32.7352, –43.7349, –22.1199, 0, 65.8548, 0.2648]
Obstacle 1	Centre /mm	(440, 0, 100)
	Half axis /mm	170, 170, 170
Obstacle 2	Centre /mm	(400, –200, 700)
	Half axis /mm	90, 80, 100
Path	Starting point /mm	(300, 435, 250)
	Target point /mm	(420, –270, 345)
System parameter of VSA-RRT	Step size **d** /°	[0.002, 0.02, 0.02, 0.02, 0.02, 0.02]
	k_{max}	10000
	Convergence precision	0.08

spend on path planning. This demonstrates that the proposed VSA-RRT algorithm can avoid the waste of sampling time caused by invalid nodes.

In order to further verify the effective of the proposed method, APF-RRT method is also used to solve this path planning problem. The simulation results by different methods are shown in Table 5.

As listed in Table 5, the sampling number is equal to the valid node number for the proposed VSA-RRT, while the sampling number is larger than the valid node number for the three methods (basic RRT, RRT with SAD and APF-RRT). This demonstrates that the basic RRT, RRT with SAD and APF-RRT will generate invalid nodes that refer to the points where the robotic arm collides with the obstacle,

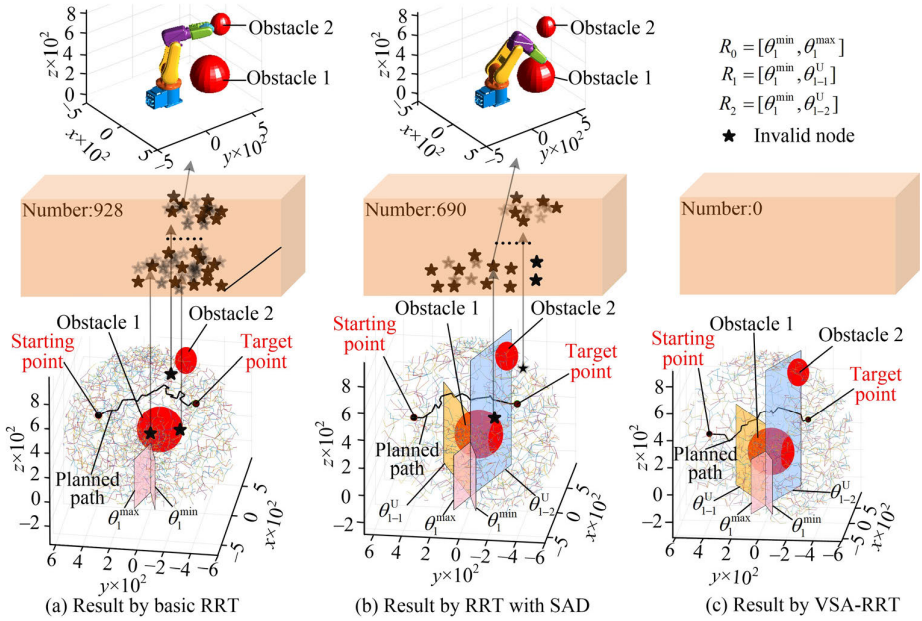

FIGURE 10. Simulation results of second example.

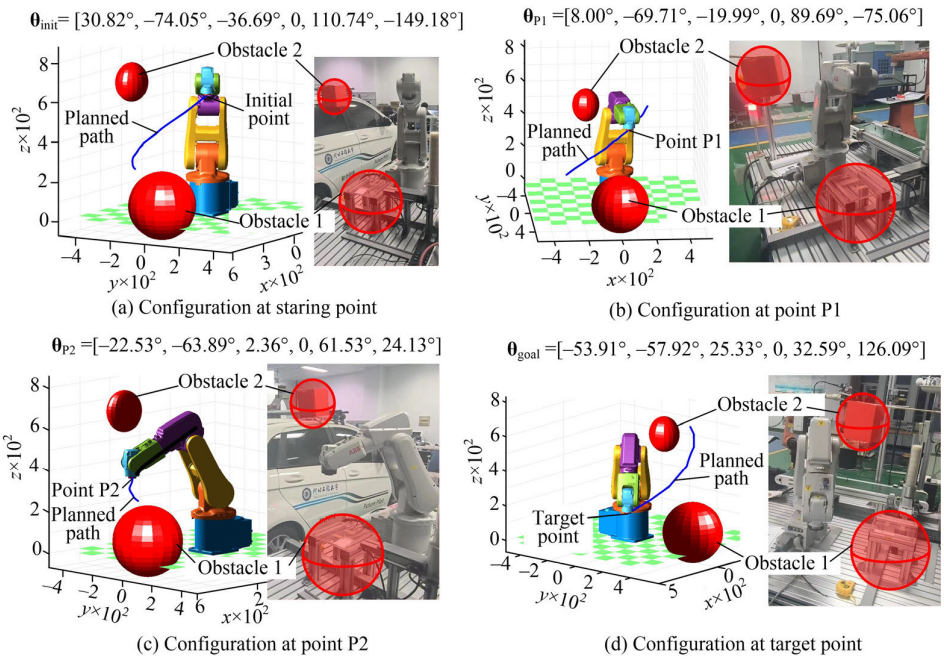

FIGURE 11. Simulation and experiment results of third example.

during their algorithm executions. Moreover, the sampling number by VSA-RRT is the least in all four methods. Compared with the basic RRT, RRT with SAD and APF-RRT, VSA-RRT decreases the sampling number by 44.98%, 19.95% and 15.00%, respectively. These results indicate that the proposed VSA-RRT has a better efficiency.

TABLE 5. Simulation results by different methods.

Method	Sampling number	Valid node number
Basic RRT	4893	3965
RRT with SAD	3363	2673
APF-RRT	3167	3049
VSA-RRT	2692	2692

C. EXPERIMENT VALIDATION

The third example is an experiment validation in real environment, which is designed to further verify the effectiveness of the proposed algorithm. Two obstacles are also designed in this validation example. The parameters of two obstacles and the system parameters of VSA-RRT are the same as the second example (see table 4). The path planning and the

joint angle simulated by VSA-RRT is used to complete the experiment. The results are shown in Fig. 11.

In Fig. 11, the configuration at starting point, target point and two process points (P1 and P2) are displayed. The coordinate of starting point, target point, point P1 and point P2 are (285, 170, 650), (215, –295, 230), (392.15, 55.11, 539.66) and (385.80, −160.04, 378.02) (Note that, the unit is mm). The initial joint angle and the target joint angle are [30.82°, −74.05°, −36.69°, 0, 110.74°, −149.18°] and [−53.91°, −57.92°, 25.33°, 0, 32.59°, 126.09°]. The joint angle of point P1 and point P2 are [8.00°, −69.71°, −19.99°, 0, 89.69°, −75.06°] and [−22.53°, −63.89°, 2.36°, 0, 61.53°, 24.13°]. As shown in Fig. 11, the planned path and the robotic arm (which is configured as the planned joint angles) both do not collide with these two obstacles. These demonstrate that the planned path and joint angles by the proposed method can be effectively applied to practical engineering.

V. CONCLUSION

In this paper, a VSA-RRT algorithm was proposed to enhance the efficiency of path planning of robotic arm. The advantages of VSA-RRT included the following two aspects. (1) VSA-RRT used the developed obstacle avoidance model to generate new sampling point, which ensured that any new sampling point is collision-free. This was mainly attributed to the obstacle avoidance model can directly recognized the collision situation with the given joint angles. (2) The sampling area dividing strategy was adopted to divide the motion space of robotic arm into several subareas, and in the sampling process of VSA-RRT, the sampling area was automatically selected from these subareas, which made the sampling process continuously move towards the target point. These two advantages benefited the efficiency improvement of VSA-RRT in path planning of robotic arm.

The three examples demonstrated the effectiveness of the developed obstacle avoidance model, sampling area dividing strategy and VSA-RRT. VSA-RRT had a better computational efficiency compared with other RRT-based path planning algorithms. However, the developed obstacle avoidance model utilized the limiting joint angle corresponding to the extreme position of the robotic arm and obstacle to achieve collision judgment, and the model only regarded the robotic arm as a motion element. This meant the working environment was regarded as unchanged, and the moving obstacles were not involved. Thus, the proposed VSA-RRT was unserviceable to dynamic environments with moving obstacles. Another limitation of VSA-RRT was that the planned path still needed post-processing before practical application, which mainly handled the smoothness of the planned path and motion stability. Therefore, future work intends to expand the obstacle avoidance model to accommodate dynamic environments with moving obstacles, and moreover, the new path planning method whose planned path can directly apply to practical applications of robotic arm is also the focus of future research.

REFERENCES

[1] L. Yang, E. Li, J. Fan, T. Long, and Z. Liang, "Automatic extraction and identification of narrow butt joint based on ANFIS before GMAW," *Int. J. Adv. Manuf. Technol.*, vol. 100, nos. 1–4, pp. 609–622, Jan. 2019.

[2] X. Wang, J. Gao, X. Zhou, and X. Gu, "Path planning for the gantry welding robot system based on improved RRT*," *Robot. Comput.-Integr. Manuf.*, vol. 85, Feb. 2024, Art. no. 102643.

[3] B. Tipary and G. Erdős, "Generic development methodology for flexible robotic pick-and-place workcells based on digital twin," *Robot. Comput.-Integr. Manuf.*, vol. 71, Oct. 2021, Art. no. 102140.

[4] C. Chen, Y. Pan, D. Li, S. Zhang, Z. Zhao, and J. Hong, "A virtual-physical collision detection interface for AR-based interactive teaching of robot," *Robot. Comput.-Integr. Manuf.*, vol. 64, Aug. 2020, Art. no. 101948.

[5] K. Kaltsoukalas, S. Makris, and G. Chryssolouris, "On generating the motion of industrial robot manipulators," *Robot. Comput.-Integr. Manuf.*, vol. 32, pp. 65–71, Apr. 2015.

[6] E. W. Dijkstra, "A note on two problems in connexion with graphs," *Numerische Math.*, vol. 1, no. 1, pp. 269–271, Dec. 1959.

[7] S. Muthuswamy and S. Manoochehri, "Optimal path planning for robot manipulators," *J. Mech. Design*, vol. 114, no. 4, pp. 586–595, Dec. 1992.

[8] Y.-A. Lu, K. Tang, and C.-Y. Wang, "Collision-free and smooth joint motion planning for six-axis industrial robots by redundancy optimization," *Robot. Comput.-Integr. Manuf.*, vol. 68, Apr. 2021, Art. no. 102091.

[9] P. Hart, N. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Trans. Syst. Sci. Cybern.*, vol. SSC-4, no. 2, pp. 100–107, Jul. 1968.

[10] P. Tavares, J. Lima, P. Costa, and A. P. Moreira, "Multiple manipulators path planning using double A," *Ind. Robot, Int. J.*, vol. 43, no. 6, pp. 657–664, Oct. 2016.

[11] L. Liu, B. Wang, and H. Xu, "Research on path-planning algorithm integrating optimization A-star algorithm and artificial potential field method," *Electronics*, vol. 11, no. 22, p. 3660, Nov. 2022.

[12] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *Int. J. Robot. Res.*, vol. 5, no. 1, pp. 90–98, Mar. 1986.

[13] U. Orozco-Rosas, O. Montiel, and R. Sepúlveda, "Mobile robot path planning using membrane evolutionary artificial potential field," *Appl. Soft Comput.*, vol. 77, pp. 236–251, Apr. 2019.

[14] O. Montiel, R. Sepúlveda, and U. Orozco-Rosas, "Optimal path planning generation for mobile robots using parallel evolutionary artificial potential field," *J. Intell. Robotic Syst.*, vol. 79, no. 2, pp. 237–257, Aug. 2015.

[15] L. E. Kavraki, P. Svestka, J.-C. Latombe, and M. H. Overmars, "Probabilistic roadmaps for path planning in high-dimensional configuration spaces," *IEEE Trans. Robot. Autom.*, vol. 12, no. 4, pp. 566–580, Aug. 1996.

[16] A. Stentz, "Optimal and efficient path planning for partially known environments," in *Proc. IEEE Int. Conf. Robot. Autom.*, Aug. 1997, vol. 388, no. 4, pp. 3310–3317.

[17] S. M. LaValle and J. J. Kuffner, "Randomized kinodynamic planning," *Int. J. Robot. Res.*, vol. 20, no. 5, pp. 378–400, May 2001.

[18] L. Larsen and J. Kim, "Path planning of cooperating industrial robots using evolutionary algorithms," *Robot. Comput.-Integr. Manuf.*, vol. 67, Feb. 2021, Art. no. 102053.

[19] L.-X. Zhang, X.-J. Meng, Z.-J. Ding, and T.-S. Wang, "Two stage path planning method for co-worked double industrial robots," *IEEE Access*, vol. 11, pp. 126995–127010, 2023.

[20] S. Wu, A. Dong, Q. Li, W. Wei, Y. Zhang, and Z. Ye, "Application of ant colony optimization algorithm based on farthest point optimization and multi-objective strategy in robot path planning," *Appl. Soft Comput.*, vol. 167, Dec. 2024, Art. no. 112433.

[21] T. Ma, S. Liu, and H. Xiao, "Multirobot searching method of natural gas leakage sources on offshore platform using ant colony optimization," *Int. J. Adv. Robotic Syst.*, vol. 17, no. 5, Sep. 2020, Art. no. 1729881420959012.

[22] Z. Zhang, B. Qiao, W. Zhao, and X. Chen, "A predictive path planning algorithm for mobile robot in dynamic environments based on rapidly exploring random tree," *Arabian J. Sci. Eng.*, vol. 46, no. 9, pp. 8223–8232, Feb. 2021.

[23] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, Jun. 2011.

[24] I.-B. Jeong, S.-J. Lee, and J.-H. Kim, "Quick-RRT: Triangular inequality-based implementation of RRT* with improved initial solution and convergence rate," *Expert Syst. Appl.*, vol. 123, pp. 82–90, Jun. 2019.

[25] B. H. Meng, I. S. Godage, and I. Kanj, "RRT-based path planning for continuum arms," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6830–6837, Jul. 2022.
[26] N. Zhang, C. Cui, and G. Wu, "Path planning of a 5-dof robotic arm based on BiRRT-APF algorithm considering obstacle avoidance," *Proc. Inst. Mech. Eng., C, J. Mech. Eng. Sci.*, vol. 236, no. 16, pp. 9282–9292, Aug. 2022.
[27] J. Wang, W. Chi, C. Li, and M. Q.-H. Meng, "Efficient robot motion planning using bidirectional-unidirectional RRT extend function," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 3, pp. 1859–1868, Jul. 2022.
[28] J. Yi, Q. Yuan, R. Sun, and H. Bai, "Path planning of a manipulator based on an improved P_RRT algorithm," *Complex Intell. Syst.*, vol. 8, no. 3, pp. 2227–2245, Jan. 2022.
[29] G. Ma, Y. Duan, M. Li, Z. Xie, and J. Zhu, "A probability smoothing bi-RRT path planning algorithm for indoor robot," *Future Gener. Comput. Syst.*, vol. 143, pp. 349–360, Jun. 2023.
[30] X. Cheng, J. Zhou, Z. Zhou, X. Zhao, J. Gao, and T. Qiao, "An improved RRT-connect path planning algorithm of robotic arm for automatic sampling of exhaust emission detection in industry 4.0," *J. Ind. Inf. Integr.*, vol. 33, Jun. 2023, Art. no. 100436.
[31] J. Ding, Y. Zhou, X. Huang, K. Song, S. Lu, and L. Wang, "An improved RRT algorithm for robot path planning based on path expansion heuristic sampling," *J. Comput. Sci.*, vol. 67, Mar. 2023, Art. no. 101937.
[32] Y. Wang, W. S. Jiang, Z. Luo, L. Yang, and Y. Q. Wang, "Path planning of a 6-DOF measuring robot with a direction guidance RRT method," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122057.
[33] Z. Feng, L. Zhou, J. Qi, and S. Hong, "DBVS-APF-RRT: A global path planning algorithm with ultra-high speed generation of initial paths and high optimal path quality," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123571.
[34] Y.-H. Yu and Y.-T. Zhang, "Collision avoidance and path planning for industrial manipulator using slice-based heuristic fast marching tree," *Robot. Comput.-Integr. Manuf.*, vol. 75, Jun. 2022, Art. no. 102289.
[35] J. Dai, Y. Zhang, and H. Deng, "Bidirectional RRT-based path planning for tight coordination of dual redundant manipulators," *Machines*, vol. 11, no. 2, p. 209, Feb. 2023.
[36] M. Kang, Q. Chen, Z. Fan, C. Yu, Y. Wang, and X. Yu, "A RRT based path planning scheme for multi-DOF robots in unstructured environments," *Comput. Electron. Agricult.*, vol. 218, Mar. 2024, Art. no. 108707.
[37] L.-X. Zhang, X.-J. Meng, and Z.-J. Ding, "Collision avoidance strategy based on virtual body deformation for path planning of serial industrial robot," *J. Mech. Sci. Technol.*, vol. 38, no. 6, pp. 3113–3129, Jun. 2024.

ZHI-JIE DING received the B.S. degree from Hebei University of Engineering, Handan, China, where he is currently pursuing the M.S. degree in mechanical engineering with the School of Mechanical and Equipment Engineering. His primary research interests include machine design, computer simulation, and industrial robot technology.

XIN-JIA MENG received the Ph.D. degree from the School of Mechanical Engineering, Beijing Institute of Technology. He is currently an Associate Professor and a M.S. Supervisor with the School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan, China. His main research interests include industrial robot technology, machine design, design optimization, and reliability-based design.

LI-XIANG ZHANG received the Ph.D. degree in mechanical manufacturing and automation from China Agricultural University, Beijing, China. She is currently an Associate Professor and a M.S. Supervisor with the School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan, China. Her main research interests include industrial robot technology, artificial intelligence, machine design, and reliability-based design optimization.

YI-BIN GUO received the B.S. degree from Hebei University of Engineering. He is currently pursuing the M.S. degree in mechanical engineering with the School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan, China. His primary research interests include industrial robotics technology, computer simulation, and artificial intelligence.

MENG-QIU CHEN received the B.S. degree from Tangshan University. He is currently pursuing the M.S. degree in mechanical engineering with the School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan, China. His primary research interests include industrial robotics technology, artificial intelligence, and computer simulation.

SHI-HANG ZHEN received the B.S. degree from Henan University of Engineering. He is currently pursuing the M.S. degree in mechanical engineering with the School of Mechanical and Equipment Engineering, Hebei University of Engineering, Handan, China. His primary research interests include industrial robotics technology, computer simulation, and artificial intelligence.